

ZE500KL Assembly Analysis

Outline

- Antenna introduction
- 2/3/4G function analysis
- WIFI & GPS function analysis

Antenna Introduction

GPS & WIFI
Antenna

Diversity
Antenna

Antenna Location

Main
Antenna
(2&3&4G)





CN Pattern
(CUCC 、 CMCC 、 WW)



TW/JP Pattern

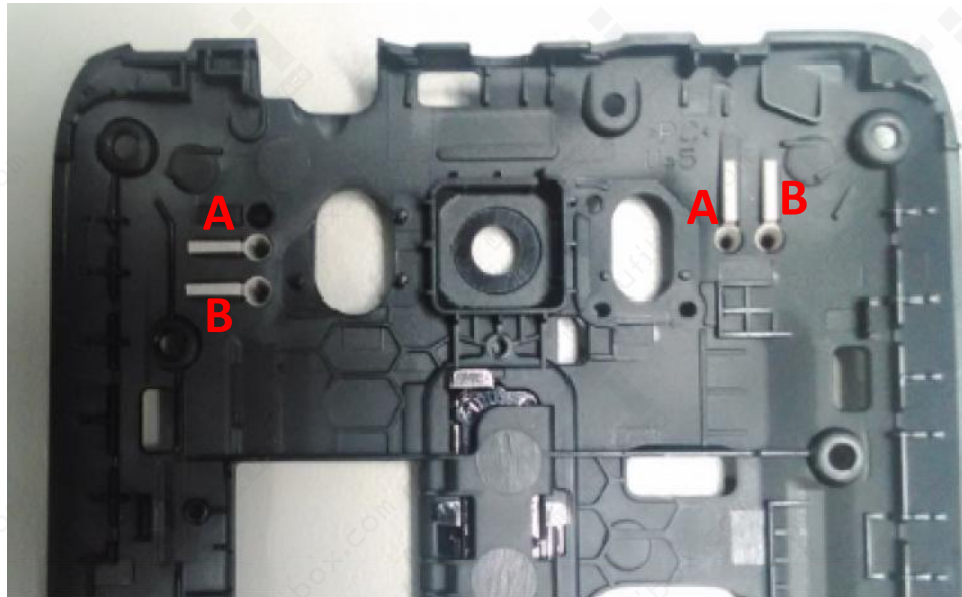


If you cannot identify the TW SKU or CN SKU, You can see from the inside of the antenna frame coding.

- * If the write-TW is the TW SKU, CN applicable to CUCC CMCC WW SKU
- * The last English letter, Q- manufacture of INPAQ, T Q- manufacture of LUXSHARE-ICT

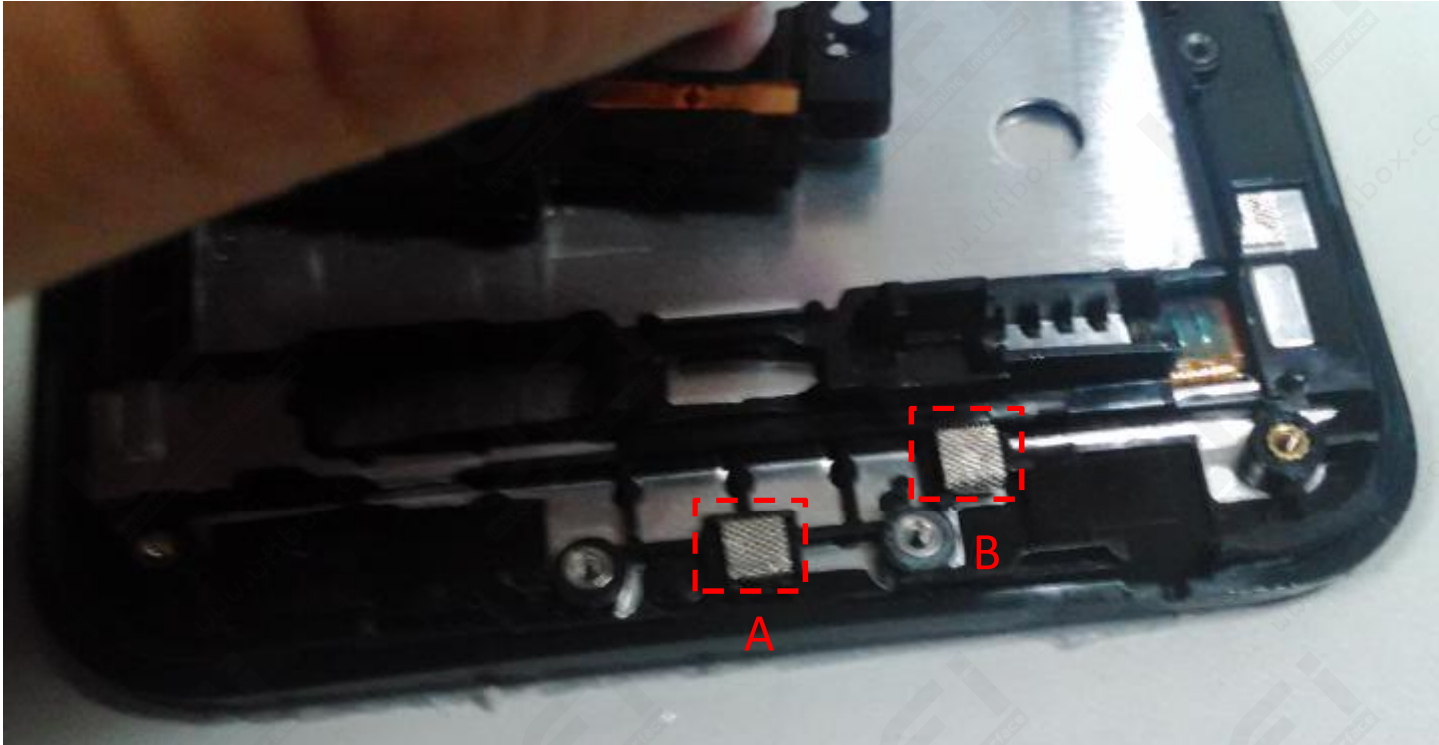
2/3/4G Quality Unhealthy **Assembly Analysis** of the Main Antenna

A. 2/3/4G Bad Signal (LDS):



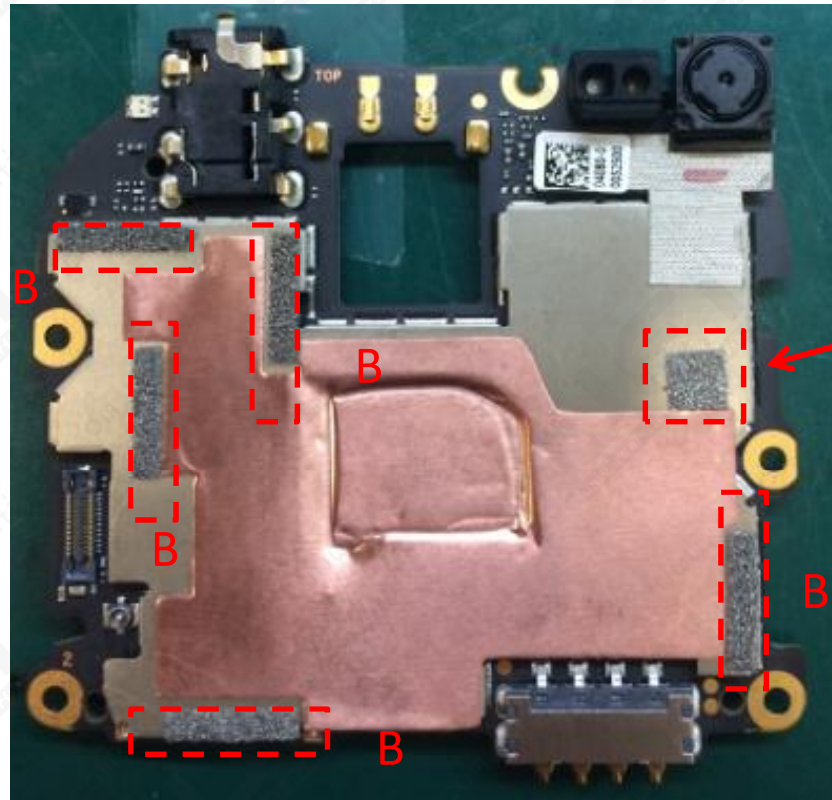
If the measured value (signal strength) have a very greatly impact, Use a voltmeter, measure each antenna at the A & B, if turn on voltage, LDS will not cut off ; If not turn on voltage, LDS will cut off
(But the probability of this happening is very low probability)

B. 2/3/4G Bad Signal(gasket):



If it is found bad call quality, you can check the sub board below the two gasket (conductive foam), No. A gasket is less impact, If the power value is too large or too small gasket is mostly caused by the No. B, Can gently press or change a new gasket.

C. 2/3/4G Bad Signal (gasket):



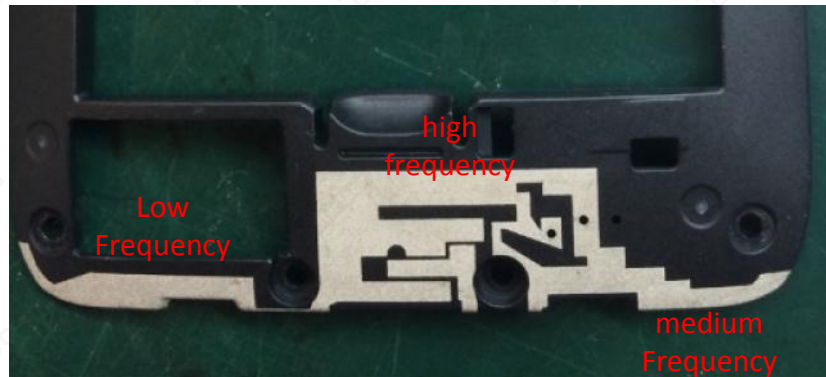
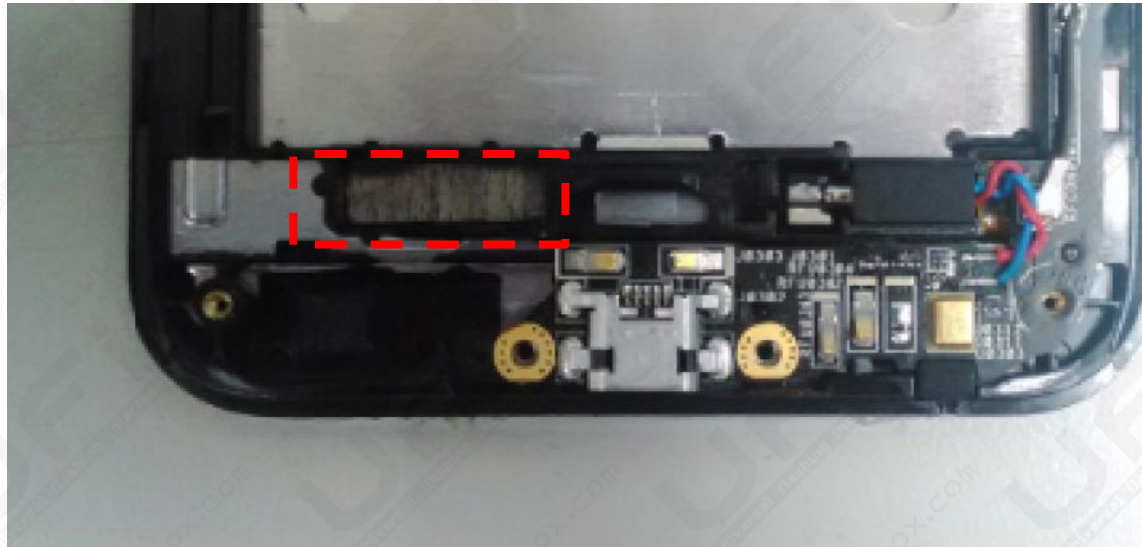
B Location : If 2/3 / 4G have poor signal, after disassemble the motherboard can be checked as illustrated conductive fabric, check all B locations conductive foam have not connected.

D. 2/3/4G Bad Signal (RF Cable):



If found power values a great change, measured values varied widely, or only half the normal, check the RF cable as shown, That should be a cable break or damage caused by the assembly.

E. 2/3/4G Bad Signal (conductive fabric docile degree):



If there are problems about the call quality, use a voltmeter to check the red mark conductive cloth is not connected properly, if there is no complete connected grounding, noise will come out, effect of low frequency RF value

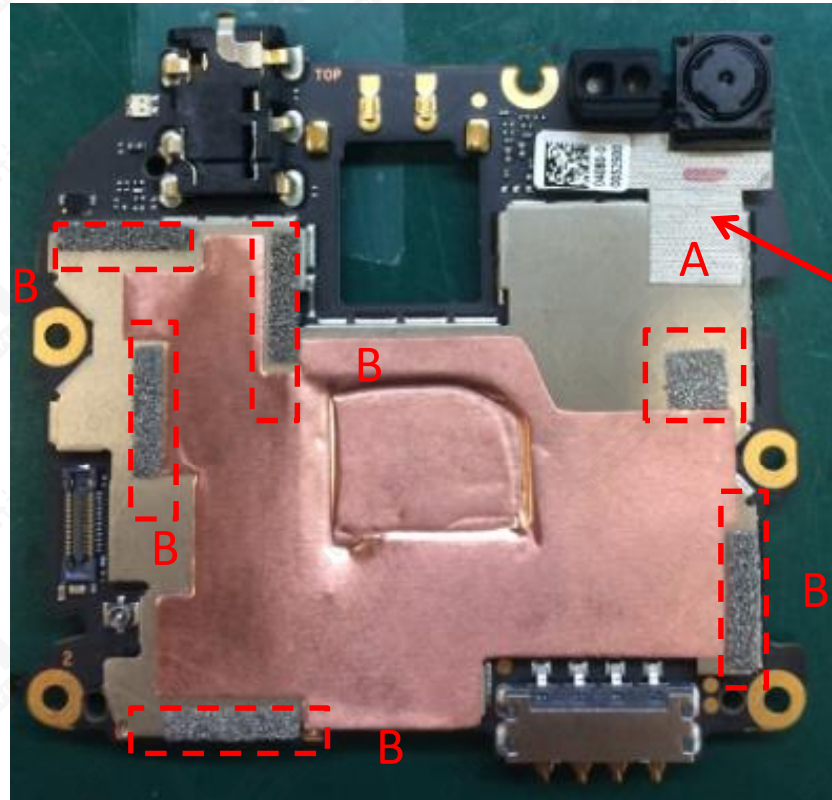
F. 2/3/4G Bad Signal (4/4):



When the machine disassembly, you can check the SPK and antennas has no trace with top of the spring, if not, it may be at risk of the contact, the impact of Power value.

WIFI /GPS Poor Quality Antenna Assembly Analysis

A. WIFI & GPS Bad Signal (BL TOP Check):

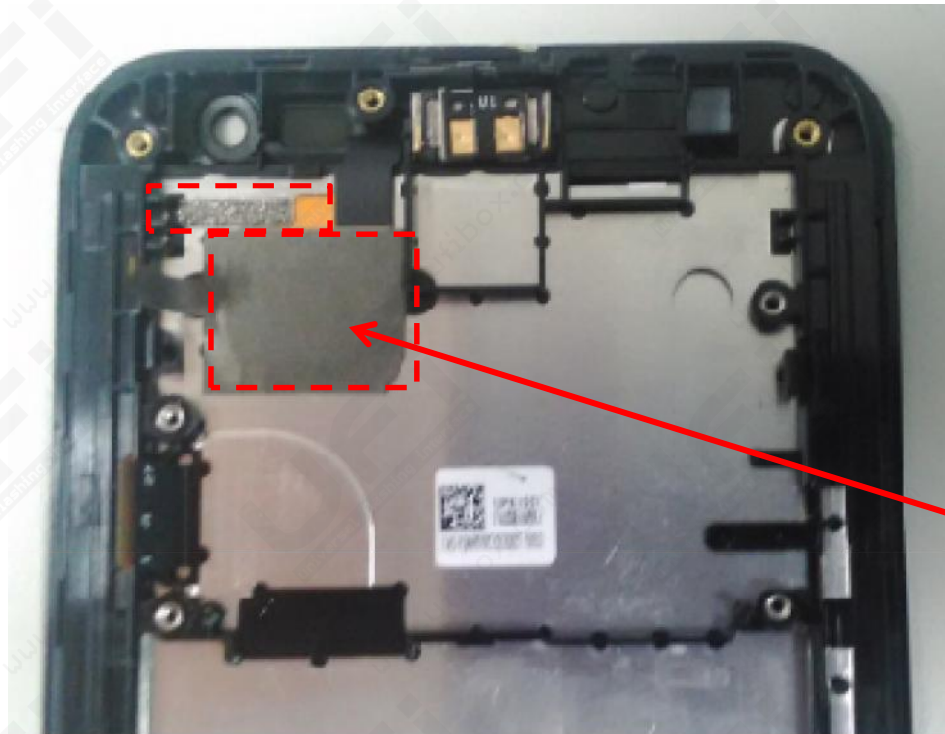


Need a good grounding

A location : If GPS or WIFI bad signal, after removing the motherboard can check the illustrated conductive foam , at A location, a large conductive foam if not connected, the GPS measurements in the production line will have little impact

B location : When the phone ON, all conductive fabric has a conduction to the whole earth, makes the phone function undisturbed.

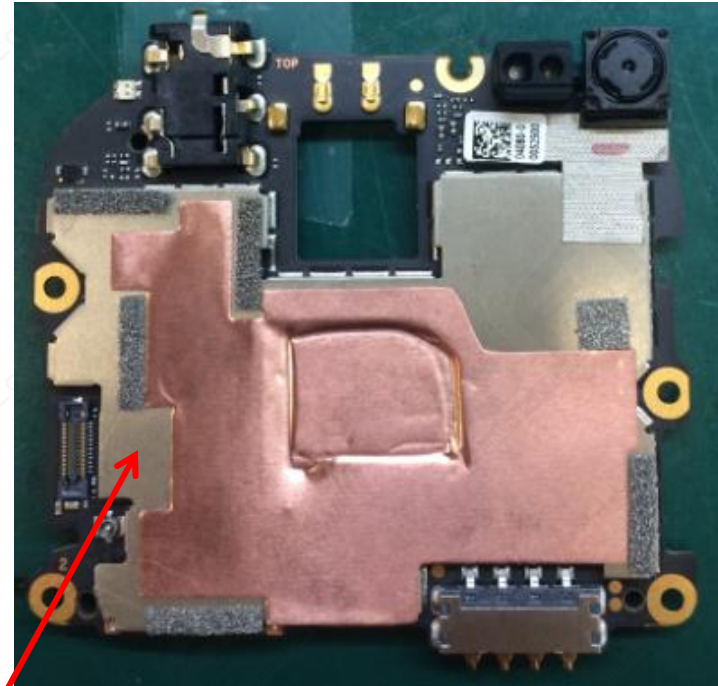
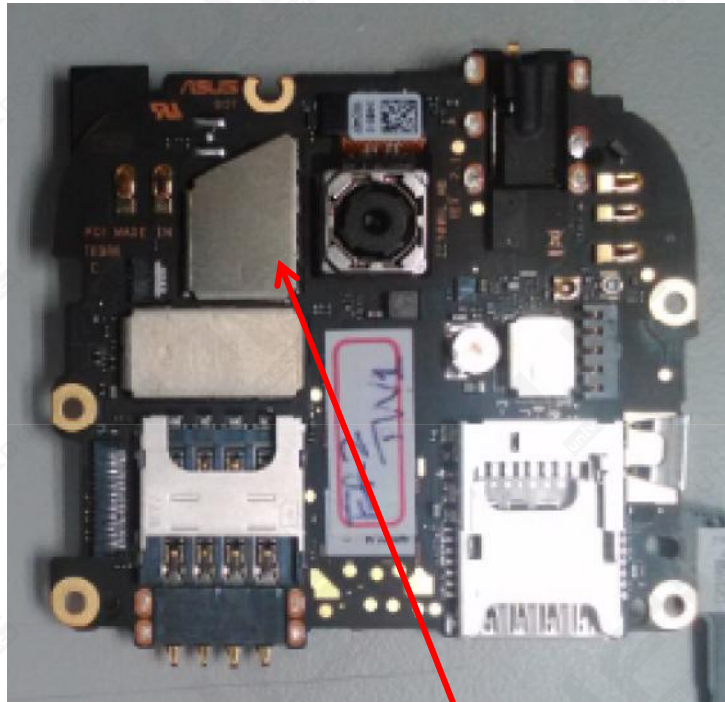
B. WIFI & GPS Bad Signal (Panel Bottom check):



Need a good grounding

If WIFI value slightly affected, you can check whether there are red frame conductive foam stickers good or flattened, if the above phenomenon, can be replaced with new foam (need a good grounding to prevent noise generation).

C. WIFI & GPS Bad Signal (Shielding case):



Shielding case

If the GPS or WIFI measured value larger gap, you can check the motherboard front and back of Shielding case there is no good soldering, this will make GPS & WIFI have a greater impact on the measurement.